

The Intersection of Data, Code, and Strategy: My Journey as an Information Professional

My name is Aaron McCullough. I am a software engineer and a final-year student at the University of South Florida, pursuing a Bachelor of Science in Information Science with a concentration in Data Science & Analytics. My journey into the world of information technology was not just sparked by a fascination with code, but by a desire to understand how data can be harnessed to solve real-world problems. Living in the Tampa Bay area, I have witnessed the rapid growth of the local tech sector, which has further fueled my ambition to become a key player in this evolving landscape. As an information professional, I view myself as a bridge between complex technical systems and the human needs they serve. This essay outlines my educational foundation, my professional experiences in software engineering and AI, and my strategic goals for a future in technical leadership.

Educational Foundation and BSIS Curriculum

My time in the BSIS program at USF has provided me with a dual-layered education that balances theoretical understanding with rigorous technical application. On the foundational level, I have explored the core tenets of Information Science, including information architecture, ethics, and policy. These courses challenged me to think critically about how information is organized, retrieved, and governed, ensuring that the systems I build are not only functional but also ethical and accessible.

Simultaneously, my concentration in Data Science & Analytics has equipped me with a specialized toolkit for the modern data economy. Through coursework in visual analytics, data mining, and statistical modeling, I have learned to extract actionable insights from raw datasets. I have engaged with the full data lifecycle—from cleaning and preprocessing to advanced visualization and predictive analysis. This academic rigor is complemented by my membership in The National Society of Leadership and Success (NSLS), which has helped me refine the soft skills necessary to communicate complex data concepts to non-technical stakeholders.

Technical Skills and Toolkit

In terms of technical proficiency, I have developed a diverse skillset that spans both software development and data analysis. My core programming competencies include Python, Java, and JavaScript (specifically React and TypeScript), which I use regularly in both my coursework and professional projects. On the data side, I am proficient in SQL for database management and have experience using R and Tableau for statistical analysis and visualization. Recently, I have focused heavily on integrating Artificial Intelligence into web applications, utilizing Large Language Models (LLMs) to create dynamic user experiences. I also hold the Google Data Analytics Professional Certificate, which served as an early catalyst for my deep dive into the field.

Professional Experience and Projects

Transitioning from theory to practice, I have been working as a Software Engineer since April 2025. In this role, I apply my academic knowledge to production-level code, tackling challenges in front-end development and system architecture. This real-world experience has been

invaluable, teaching me the importance of agile methodologies, version control, and collaborative debugging.

Beyond my employment, I am an active builder in the "indie hacker" space. I am currently developing "Budget BOOM," an AI-powered budgeting application designed to help users manage their finances through intelligent automation. This project has allowed me to wear multiple hats—from product designer to full-stack developer—and has deepened my understanding of the product lifecycle. Additionally, I write "The AI Debrief," a newsletter where I curate and analyze the latest developments in artificial intelligence. This initiative not only keeps me on the cutting edge of industry trends but also serves as a platform to practice technical writing and community engagement.

Career Goals and Future Outlook

Looking ahead, my career goals are centered on the convergence of software engineering and strategic leadership. Upon graduation in Spring 2026, I plan to continue advancing in my current trajectory as a software engineer, with a specific focus on AI integration and data-driven product development. In the short term, I aim to ship "Budget BOOM" to a wider user base and perhaps seek a specialized certification in cloud computing (such as AWS Solutions Architect) to bolster my infrastructure skills.

Long-term, I envision myself transitioning into a technical leadership role, such as a Lead Engineer or Product Manager, where I can guide teams in building scalable, impactful solutions. To prepare for this transition, I have begun researching MBA programs, specifically targeting top-tier institutions that offer a strong focus on technology management. I believe that combining my deep technical background with advanced business acumen will position me to drive innovation effectively. Ultimately, my goal is to remain a lifelong learner, continuously adapting to the shifting sands of the technology landscape while delivering value as a principled information professional.

About Me / Portfolio

To view samples of my code, data visualizations, and project breakdowns, please visit my portfolio at: <https://aaronm0710.github.io/portfolio-website/>